Construction and installation instructions for the FreeVario V2

Table of Contents

I. General..... 2

II. Assembling the boards and assembling the vario gauge..... 2

III. Programming and updating the two ESP32.....6

IV. Backing up and restoring your settings.....11

V. Install the new OV image including XCSoar with FreeVario driver.....12

VI. Install XCSoar incl. FreeVario driver without reinstalling the OV image..... 13

VII. Switch off the vario sound from OpenVario.....13

VIII. Settings for a doubleseater.....14

IX. Blocking the standard profile for club use..... 14

X. Settings in XCSoar.....15

XI. Settings for the stick remote control with FreeVario.....15

XII. Settings for the stick remote control without FreeVario.....17

XIII. Installation of the new OpenVario menu from Kedder.....18

XIV. Update maps, airspaces etc. with the new menu from Kedder.....18

XV. Using an LTE stick.....18

XVI. Installation..... 19

XVII. Disclaimer..... 20

Rev. 2.3 (06.2026)

I work on a Mac or Linux system. Little things can differ on a Windows computer!

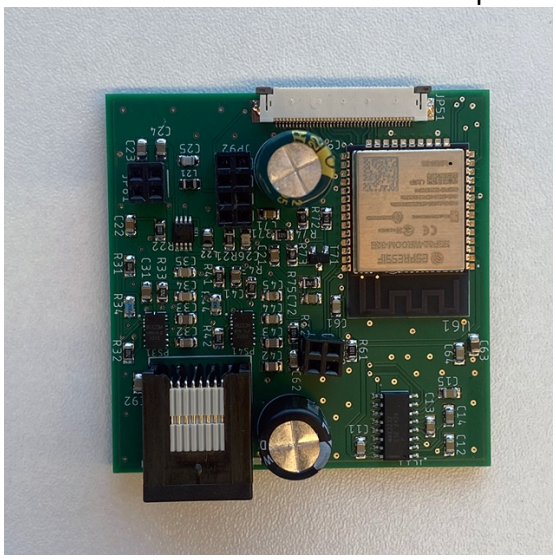
Copyright Dirk Jung licensed under CC BY-SA

I. General

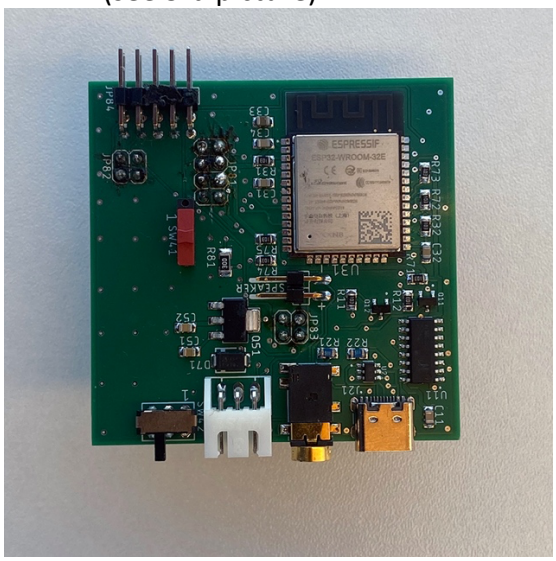
The FreeVario is operated with 12V, but can also alternatively be operated with 5V.
The dimensional accuracy of the 3D printed parts is extremely dependent on the settings of your printer and the material. So it can happen that the parts are too small. Then you have to print the parts a few percent larger.

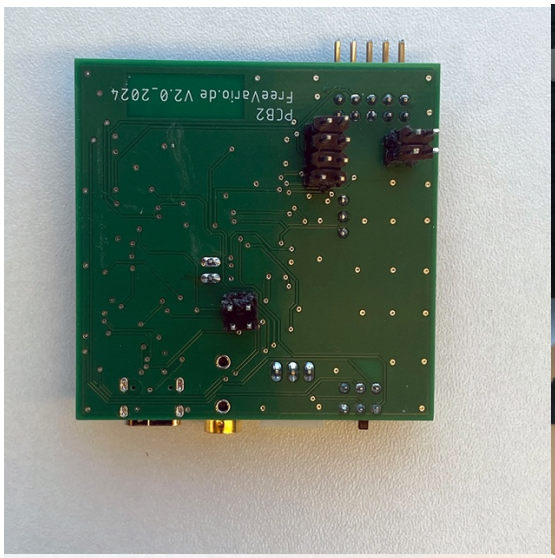
II. Assembling the boards and assembling the vario gauge

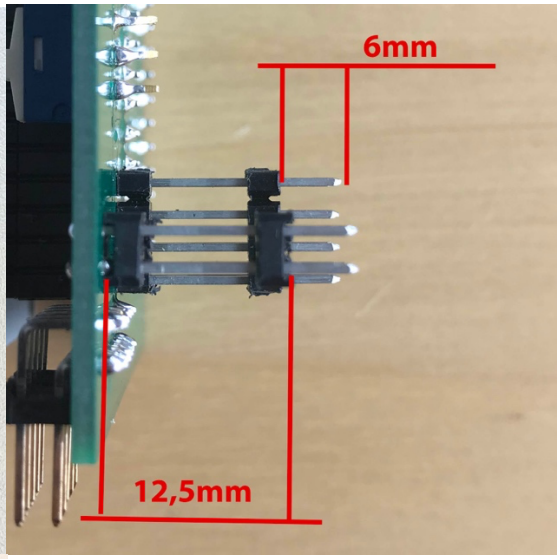
1. Solder PCB1 as well as shown in the pictures.

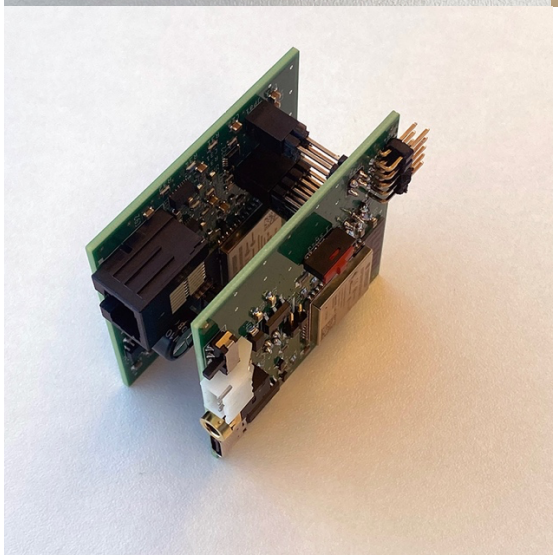


2. Solder PCB2 together as shown in the pictures. The distance between the two plastic brackets of the stacking bar must be set to 8mm. Shorten the pins that are too long to 22 mm (see 3rd picture).







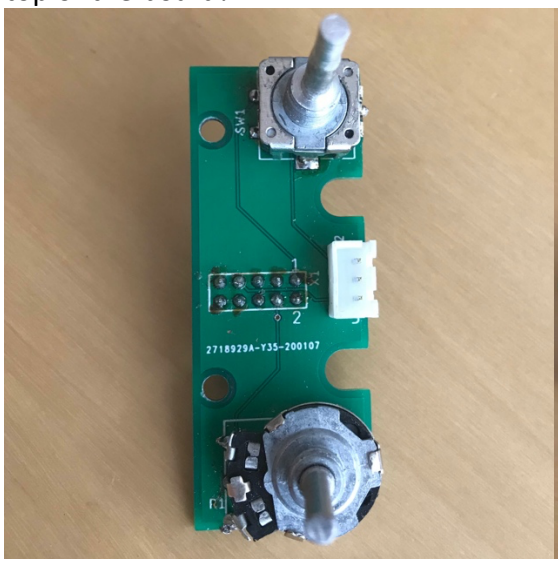


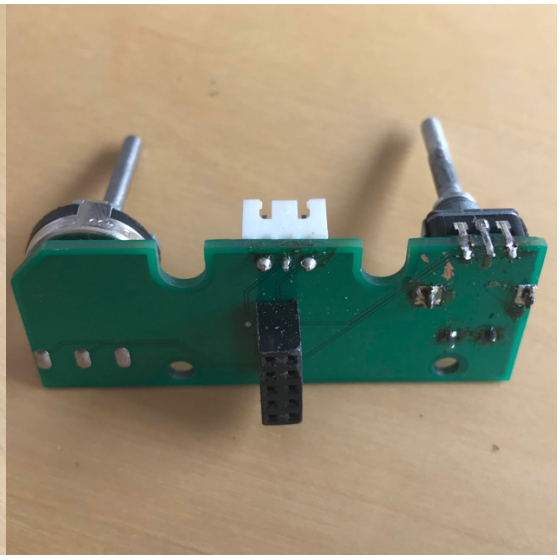
The 3.5mm jack socket is used to connect an external speaker. If the external speaker is connected, the internal one is automatically switched off.

3. Carefully disassemble the rotary encoder by bending the small metal brackets. The axis of the rotary encoder is secured with a locking ring. You have to go carefully with a side cutter into the gap between the axis and the upper part of the housing and press lightly. If the axis is slightly damaged by the side cutter, file away the spot. Clamp the axes in the lathe, turn by 4mm. Then carefully reassemble the rotary encoder and bend the metal brackets back. Secure the axis of the rotary encoder again with the ring. For safety's sake, buy the encoder twice, the first time it may go wrong. Drills the banjo bolts to 4.3mm.

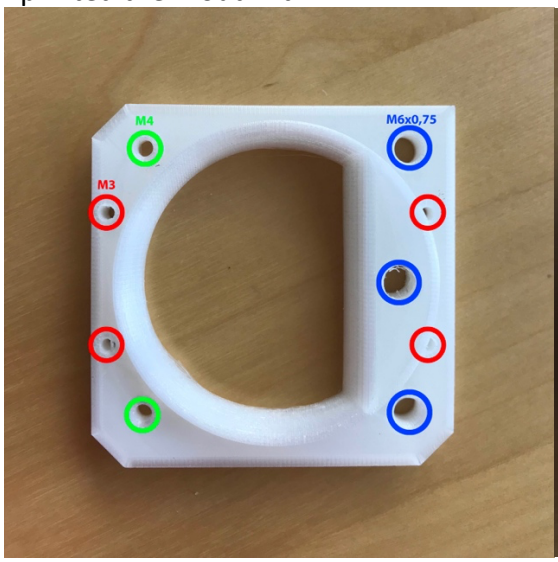
4. On the PCB3 at the position where the encoder is located, you have to file away the protruding part of the board on the side with the 3 PINs and above, otherwise the board will not fit into the housing without tension. It may well be that you have to file a little more after soldering. Carefully bend the contacts of the potentiometer downwards by 90° and solder them on. Tests whether the axes of the encoder and the potentiometer fit well through the banjo bolts and are easy to operate after installing the front plate. If necessary, loosen again and move something.Solder PCB3 as well as

shown in the pictures. Make sure that you solder the white connector so that the PINs are on the right side. Otherwise the board will not fit into the housing later. Shorten the axes to a length of 34mm + thickness of the instrument board, measured from the top of the board!

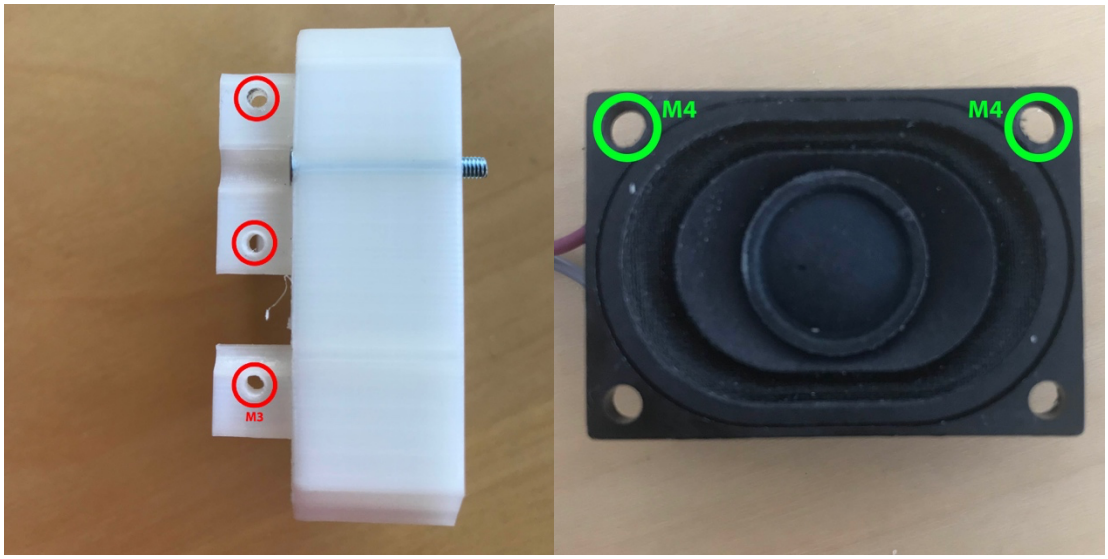




5. Print out the three housing parts and cut the threads into the plastic as marked in the pictures. Red = M3, green = M4, blue = M6x0.75.
I printed them out with PLA.







6. Cut two M4 threads in the speaker and fasten the speaker in the housing with 2 countersunk screws M4x8. Solder about 10cm long cables to connect the speaker beforehand. **The speaker is only required if the FreeVario is to output the vario sound. If you want to continue using the vario sound from OpenVario, you can omit the speaker.**

7. Crimp the connector to an approximately 5cm long, 3-pin cable and solder it to the toggle switch. Solder as space-saving as possible by soldering at an angle. Otherwise there will be a short circuit later between the solder joints and the contacts behind them on the PCB3. Make shrink wise the solder contacts!
   The middle cable comes to the middle solder lug. Depending on how you solder the two outer cables, you will later have STF or Vario on top. If you solder the upper cable of the plug to the upper soldering tab of the toggle switch, and the lower cable of the plug to the lower soldering tab of the toggle switch, STF will be in the lower position of the toggle switch, and Vario in the upper. Automatic is in the middle.

8. Shorten the threads of the two M4 fastening screws to 3mm + instrument panel thickness. Work very precisely here, because if the screws are too long, press the display later and destroy it. It is best to screw the front panel into the cockpit on your own and check that the screws do not protrude into the area of the display.

9. Install the toggle switch and the display in the front panel.

10. Insert PCB3 in the middle part and screw tight with 2 screws M3x6. Tighten screws carefully, the threads are plastic!

11. Screw the front panel and middle section together using 4 countersunk screws M3x25. Connect the toggle switch beforehand and lead the FFC cable back through the slot. Tighten screws carefully, the threads are in plastic!

12. Now first test whether the two circuit boards can be easily pushed into the pot individually. Otherwise damage can occur later. If in doubt, rework the groove a little.

13. Carefully assemble PCB1 and PCB2, connect display, loudspeaker (do not interchange plus and minus) and connect PBC3. Close the clamping mechanism of the display cable very carefully with a small screwdriver or something similar. If you slip with your fingers or the wrong tool, it is easy for you to tear the ribbon cable.

14. The best thing to do now is to jump to the programming of the two ESP32 and then continue here.

15. Now push the housing pot carefully over the circuit boards. Make sure that the boards are in the rails and carefully guide the Western Digital socket through the rear wall.

16. Screw the housing parts together with 6 countersunk screws M3x6. Tighten screws carefully, the threads are in plastic!

17. Using the two fastening screws and the two banjo screws, install the Vario in the I-board, mount the knobs and connect the Vario. Tighten screws carefully, the threads are in plastic!

18. The connection cable has the same assignment on both sides, i.e. same color on the same PIN, like a normal LAN cable.

III. Programming and updating the two ESP32

With the small slide switch on the back of the FreeVario you can choose which of the two ESPs you want to describe (D=Display-ESP, S=Sound-ESP).

Variant 1: the software version of the FreeVario is lower than 1.2 or the ESPs have not yet been programmed:

1. Put the Binaries folder e.g. to the desktop

2. Download ALL the binaries from the website and save them in this folder

3. Open a terminal or the command prompt. There you can see if you can find the path to Binaries. For example, it is /Users/PC1/Desktop/Binaries/

4. In Arduino IDE click on Arduino -> settings.

5. Add the URL under "Additional Board Administrator URLs". If URLs have already been entered, add further URLs with commas and spaces.
   https://dl.espressif.com/dl/package_esp32_index.json

6. Also click here under "Verbose output during" "Upload".

7. Click Tools -> Board -> Board Administrator and search for ESP32.

8. Select „ESP32 by Espressif Systems“ (max. till version 2.0.17) and install.

9. You may have to install the driver for the serial adapter of the ESP. https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers

10. Open and save a new sketch. Remember the name. To upload, set the following under Tools:
    - Board: DOIT ESP32 DEVKIT V1
    - Upload Speed: 115200
    - Flash Frequency: 80MHz
    - Core Debug Level: “none”
    - Port: Check out which is added after connecting the ESP32 with the USB cable

11. Then upload the empty sketch. There is a very long command in white letters above the error message in the Arduino IDE, which is roughly as follows:
    /Users/PC1/Documents/Arduino/hardware/espressif/esp32/tools/esptool/esptool --chip esp32 --port /dev/cu.SLAB_USBtoUART --baud 115200 --before default_reset --after hard_reset write_flash -z --flash_mode dio --flash_freq 80m --flash_size detect 0xe000
    /Users/PC1/Documents/Arduino/hardware/espressif/esp32/tools/partitions/boot_app0.bin 0x1000
    /Users/PC1/Documents/Arduino/hardware/espressif/esp32/tools/sdk/bin/bootloader_dio_80m.bin 0x10000
    /var/folders/6t/qjgfw2413f7ddnjzk_08btxw0000gp/T/arduino_build_661147/sketch_dec30d.ino.bin 0x8000
    /var/folders/6t/qjgfw2413f7ddnjzk_08btxw0000gp/T/arduino_build_661147/sketch_dec30d.ino.partitions.bin

12. Copy this command, you need it to load the file onto the ESP32.

13. The first part of the command shows you where Arduino IDE creates its working directory. Remember the place. In this example it would be:
    /Users/PC1/Documents/Arduino

14. Change the command with your location where the Binaries folder is located. Also change the last two file names (see bold).

    Display-ESP:
    /Users/PC1/Documents/Arduino/hardware/espressif/esp32/tools/esptool/esptool --chip esp32 --port /dev/cu.SLAB_USBtoUART --baud 115200 --before default_reset --after hard_reset write_flash -z --flash_mode dio --flash_freq 80m --flash_size detect 0xe000 ***/Users/PC1/Desktop/Binaries****/boot_app0.bin 0x1000*
    /Users/PC1/Desktop/Binaries/FreeVarioGauge.ino.bootloader.bin *0x10000*
    /Users/PC1/Desktop/Binaries/FreeVarioGauge.ino.bin *0x8000*
    /Users/PC1/Desktop/Binaries/FreeVarioGauge.ino.partitions.bin

    Sound-ESP:
    /Users/PC1/Documents/Arduino/hardware/espressif/esp32/tools/esptool/esptool --chip esp32 --port /dev/cu.SLAB_USBtoUART --baud 115200 --before default_reset --

after hard_reset write_flash -z --flash_mode dio --flash_freq 80m --flash_size detect 0xe000 ***/Users/PC1/Desktop/Binaries****/boot_app0.bin 0x1000*
/Users/PC1/Desktop/Binaries/VarioSound.ino.bootloader.bin *0x10000*
/Users/PC1/Desktop/Binaries/VarioSound.ino.bin *0x8000*
/Users/PC1/Desktop/Binaries/VarioSound.ino.partitions.bin

15. Connect the ESP32 to the PC using the USB cable and open a terminal or the command prompt. Issue the **appropriate** command there.

16. After the upload is complete, the ESP32 restarts. The files then have to be uploaded to the memory of the respective ESP32.

17. The ESP32 Sketch Data Upload program currently only works up to Ardoino IDE 1.8.19!

18. Install ESP32 Sketch Data Upload. Download ESP32FS-1.1.zip from https://github.com/me-no-dev/arduino-esp32fs-plugin/releases/

19. On Mac, copy the extracted folder ESP32FS to /Documents/Arduino/tools.

20. Using Windows, copy the extracted folder ESP32FS to /Programs/Arduino/tools.

21. Download and unzip the sound_data.zip and display_data.zip files from the website. Go to the working directory of the Arduino IDE (e.g. /Users/PC1/Documents/Arduino) and there to the folder of the sketch you just created. Copy the unzipped data folder for the sound ESP or the display ESP here, depending on which one you want to upload to

22. Restart the Arduino IDE and open your newly created sketch.

23. IMPORTANT!! Make sure the serial monitor is closed.

24. Click on “ESP32 Sketch Data Upload” under Tools.

25. After the upload is complete, the ESP32 restarts and, if the display is already connected, you can now see the display of the vario gauge.

26. Now go back to point 13 of the assembly part of the instructions.

Variant 2: the software version of the FreeVario is at least 1.2:

Registering the ESPs in a WLAN

1. Switch on the FreeVario and, if possible, leave the OpenVario off. If it is not possible to switch on the devices independently of each other, end via the menu in OpenVario XCSoar. XCSoar should not try to communicate with the FreeVario during the update.

2. As long as the display shows the version number, you can switch to update mode.

3. To do this, move the toggle switch from the top to the bottom position, or from the bottom to the top position.

4. The soundboard acknowledges the update mode with a peep tone, the displayboard shows "starting update mode" on the screen.

5. If the two ESPs do not find a WLAN to which they can connect, start in access point mode, which means they provide a WLAN themselves.

6. As it takes a while to try to connect to a wireless network, this process may take around 1 minute.

7. What mode each ESP is in and what IP it can be reached on is displayed on the screen.

8. The access point of the soundboard is called "FreeVario_Soundboard", the access point of the displayboard is called "FreeVario_Displayboard"

9. Connect to one of the two access points with your computer or mobile phone and call up the respective IP displayed in the browser.

10. There you will find a list of the WiFi networks found. Choose the one you want the ESP to connect to and enter the appropriate SSID and WiFi password.

11. Now the connection data is saved in the EEPROM. Only the connection data for a WLAN can be saved there. If you connect to someone else, the old connection data will be deleted.

12. Restart the FreeVario, go back to update mode and repeat steps 9 - 11 with the other access point.

13. You don't have to repeat steps 1 - 12 in the future if you stay in the same WLAN.

ESPs already connect to a WLAN

14. Download the binary files VarioSound.ino.bin and FreeVarioGauge.ino.bin from FreeVario.de and save them on your device.

15. Restart the FreeVario and return to update mode.

16. Now both ESPs connect to the selected WLAN. Connect your computer or mobile phone to this WLAN as well and call up one of the two IPs shown on the display in the browser.

17. Enter "Sound" for the password and login on the Soundboard.

18. Enter "Display" for the password and login on the display board.

19. Select the VarioSound.ino.bin file for the Soundboard and the FreeVarioGauge.ino.bin file for the Displayboard and click on Update.

20. The update takes about 1 minute and the successful completion is indicated by a restart of the display on the display board. Unfortunately, there is no feedback from the soundboard.

21. Restart the FreeVario, go back to update mode and repeat steps 16 - 20 for the other ESP.

22. The process is complete after restarting the FreeVario.

Variant 3: the software version of the FreeVario is at least 1.2.1:

1. Download the binary files VarioSound.ino.bin and FreeVarioGauge.ino.bin from FreeVario.de and save them on your device.

2. Switch on the FreeVario and, if possible, leave the OpenVario off. If it is not possible to switch on the devices independently of each other, end via the menu in OpenVario XCSoar. XCSoar should not try to communicate with the FreeVario during the update.

3. As long as the display shows the version number, you can switch to update mode.

4. To do this, move the toggle switch from the top to the bottom position, or from the bottom to the top position.

5. The soundboard acknowledges the update mode with a peep tone, the displayboard shows "starting update mode" on the screen.

6. Now the display board provides the WLAN FV_Displayboard and the soundboard provides the WLAN FV_Soundboard. The password is 12345678 in each case.

7. The ESP32's WLAN generates noise in the audio amplifier, this is unfortunately normal.

8. If you have connected to one of the two WLANs, you can call up the update page of the respective board in your browser.

9. Enter FreeVario_Displayboard.local or FreeVario_Soundboard.local as the URL, or alternatively the IP shown on the display.

10. Select the VarioSound.ino.bin file for the Soundboard and the FreeVarioGauge.ino.bin file for the Displayboard and click on Update.

11. The update takes about 1 minute and successful completion is indicated by a restart of the display on the display board. Unfortunately, there is no feedback from the soundboard.

12. Restart the FreeVario, go back to update mode and repeat steps 16 - 20 for the other ESP.

13. The process is complete after restarting the FreeVario.

IV. Backing up and restoring your settings

1. Save the settings:

Variant 1: the original menu is used:

- Back up the .xcsoar folder from the old system
- Boot OpenVario and go to the menu. Select "Exit to the shell" and confirm with Yes. Connect your USB stick to manage the OpenVario and a USB keyboard. Enter the following commands.
- mkdir /tmp/USB
- mount /dev/„USB-Stick“ /tmp/USB (/„USB-Stick“ usually is /dev/sda1)
- cp -r .xcsoar /tmp/USB (takes a bit of time!)
- cd /tmp/USB
- mv .xcsoar xcsoar
- rm -r openvario/upload/xcsoar
- mv xcsoar openvario/upload/
- Save the value for calibrating the voltage display (if you have calibrated it at all, skip step otherwise):
- Boot OpenVario and go to the menu. Select "Exit to the shell" and confirm with Yes. Connect a USB keyboard and enter the following commands.
- nano /opt/conf/sensord.conf
- write down the value for the "voltage_config" variable. For me it was „voltage_config 1592.4 4.54545“.
- 1592.4 is the scaling factor, 4.54545 is the offset. The formula to convert the value of the AD converter into the voltage value is:
- U = (value AD converter / scaling factor) + offset

- Save the WiFi settings:
- Copy the folder /var/lib/connman to the USB stick, which is already mounted.
- cp -r /var/lib/connman /tmp/USB

Variant 2: the new menu from Kedder is used:

- Insert an empty FAT32 formatted USB stick into the OpenVario
- Start the OpenVario and select “Applications” -> “Backup” -> “Backup” in the menu
- Everything important is now automatically backed up

2. Upload your saved data to the OpenVario:

Variant 1: the original menu is used:

- Insert the new SD card into the OpenVario and start. Then go back to the OpenVario menu.
- Click "Copy file to and from OpenVario" -> "Upload files from USB to XCSoar". This will restore all of your settings. Wait until “done !!” is shown.
- Then set the desired orientation of the display under "Update, Settings, ..." -> "System Settings" with "Set rotation of the display".
- If necessary, select “Calibrate Touch” under “Update, Settings,…”.
- Select “Calibrate Sensors” under “Update, Settings,…”.

- Finally, set the language under "Update, Settings, ..." -> "System Settings" -> "Set language used for XCSoar".
- Back to the main menu with ESC and select "Restart" and confirm with Yes.

- Restore the voltage display calibration:
- nano /opt/conf/sensord.conf
- „voltage_config" reset to the noted values

- Restore backup of old WiFi settings:
- Copy the connman folder saved on the USB stick back into the /var/lib directory
- mkdir /tmp/USB
- mount /dev/„USB-Stick" /tmp/USB (/„USB-Stick" usually is /dev/sda1
- cp -r /tmp/USB/connman /var/lib

- Set up WiFi again if there is no backup of the old settings:
- Boot OpenVario and go to the menu. Select "Exit to the shell" and confirm with Yes. Connect a USB keyboard and enter the following commands.
- connmanctl
- enable wifi
- scan wifi
- services
- agent on
- connect wifi_...
- Enter password

Variant 2: the new menu from Kedder is used:
- Insert the USB stick with your backup into the OpenVario
- Start the OpenVario and select "Applications" -> "Backup" -> "Restore" in the menu
- Everything important is now automatically restored

V. Install the new OV image including XCSoar with FreeVario driver

1. From the official Image 21033 of OpenVario, the FreeVario driver is integrated in the image. So you are no longer dependent on using images from FreeVario.de. If you have already installed an image higher the 21032, you can connect the FreeVario directly to your OpenVario without making any changes at the software.

2. Install the new image:

Variant 1: Use a new SD card (secure method)
- Use a Linux PC to write the new image to another SD card. So you can be sure that on the old map there is still a working system in case something goes wrong.
- unzip the image first:
- gunzip "Path_to_Image"
- e.g .: gunzip /home/USER/Schreibtisch/OpenVario-linux-openvario-image-testing-glibc-ipk-20149-openvario-7-PQ070.rootfs.img.gz

- dd if=“ Path_to_image“ of=/dev/„ new_SD card “
- z.B.: dd if=OpenVario_new.img of=/dev/sda

Variant 2: Overwrite old SD card (insecure method)

- In the main directory of the USB stick "openvario" you put the recovery file "ov-recovery.itb", which you can download on the FTP server (ftp://ftp.openvario.org/recovery).
- IMPORTANT!! If you do not want to perform a recovery, the file "ov-recovery.itb" must be renamed, e.g. in "ov-recovery.xxx".
- Copie the image to be installed as a .gz file to the USB stick in the images subfolder
- Insert the USB stick into the OpenVario and boot.
- The recovery menu is shown in red. Remember that all files in OpenVario will be deleted when you restore!
- Select "Write image to SD Card" and then "Update complete SD Card".

VI. Install XCSoar incl. FreeVario driver without reinstalling the OV image

1. If you don't want to reinstall the complete image of OV, but only want to replace the version of XCSoar, you can do that as described below. Your settings in XCSoar remain completely intact. I would always create a backup file of the SD card beforehand. Something always can go wrong!

2. Copy the new version of the installation file from XCSoar to a USB stick, select "Exit to the shell" in the OpenVario menu. Find the path of the USB stick with fdisk -l (e.g. /dev/sda1).

3. mkdir /tmp/USB

4. mount /dev/sda1 /tmp/USB

5. cd /tmp/USB

6. opkg remove xcsoar

7. opkg install xcsoar*.ipk

8. Set the language in the OpenVario menu again and reboot.

VII. Switch off the vario sound from OpenVario

1. **Skip this section completely if you want to continue using the vario sound from OpenVario. If you wanted the FreeVario to make the sound, you have to work through this section.**

2. Choose "Exit to the shell" from the OpenVario menu and enter the command "systemctl disable variod" to switch off the vario sound from OpenVario.

3. Reboot OpenVario.

4. Changes the port of NMEA port A from 4352 to 4353.

5. If vario sound can still be heard in XCSoar, it will be generated by XCSoar.

6. At System -> Display -> Audio Vario set Audio Vario to ON and set volume at 0%. Confirm with Close.

VIII. Settings for a doubleseater

1. To connect the front and rear OpenVario to each other, you have to use a crossed cable. That means you have to swap PINs 5 and 6 for RX and TX on one connector.

2. The front device is set to the RJ45 socket to be used, e.g. ttyS3, NMEA output is set as driver, the baud rate e.g. to 38400 (it must be the same in front and in the back!)

3. The rear device is also set to the RJ45 socket to be used, e.g. ttyS3, OpenVario is set as the driver, and the baud rate must be set in the same way as the front device.

IX. Blocking the standard profile for club use

1. Go to the console in the main menu of the OpenVario, in which you select "Exit to Shell".

2. Go to the directory in which the profiles are saved with "cd .xcsoar".

3. By default, the profile is called openvario.prf. I renamed it to clubprofile.prf (mv openvario.prf clubprofile.prf)

4. Lock the profile with the following command:
   chattr +i clubprofile.prf

5. Unlock the profile with the following command:
   chattr -i clubprofile.prf

X. Settings in XCSoar

1. Click Config. -> Devices, select a free device from A, B, C or D and select Edit.

2. You should assign the connections so that the Flarm is above the FreeVario.

3. Select the port, where the electronical vario is connected with (e.g. ttyS1).

4. Baud rate is 115200, Driver is FreeVario, confirm with OK and Close.

XI. Settings for the stick remote control with FreeVario

In order to be able to use the STF switch to control the FreeVario in automatic mode, a few settings must be made. The changeover from Vario to STF, or vice versa, then works as follows:

1. priority has the toggle switch of the FreeVario. If STF or Vario is selected with it, the FreeVario is set to the respective mode. If the toggle switch of the FreeVario is on automatic, the 2nd priority comes into play.
2. The stick remote control has priority. A short click activates Vario, a long click STF and a double click the automatic mode. The mode set with the stick remote control is sent to the FreeVario and to the OpenVario to change the tone. If the stick remote control is also in automatic mode, the third priority is active.
3. Depending on which option was selected in the menu under STF, priority is now given to XCSoar or a switch connected to the flaps. Here, too, only the FreeVario is being switched at the moment. A command to the OpenVario to switch the sound does not work at the moment.

Important!! Sets in XCSoar OpenVario on device A and FreeVario on device B. Otherwise it won't work !!

1. The image of the FreeVario page must be newer than 22028 on the OpenVario. How to do this is given in Chapter V.

2. On the FreeVario, the ESPs of the display board and the soundboard must be updated to at least version 1.1.4. How to do this is in Chapter III.

3. If you use the openvario.xci file originally from the image and have not changed it to sort the menu or anything similar, you don't need to do anything here. Be careful, if you import a backup, the openvario.xci file will automatically be overwritten with your old version. Then you should first change the file in your backup with a text editor. The text editor is not allowed to do any text formatting, I use VSCode for that. If you use your own file, you must also make the following changes:

```
mode=default
type=key
data=V
```

```
event=SendNMEAPort1 POV,C,VAR
event=StatusMessage Vario Mode

mode=default
type=key
data=S
event=SendNMEAPort1 POV,C,STF
event=StatusMessage Speed to Fly Mode
```

Change to:

```
mode=default
type=key
data=V
#event=SendNMEAPort1 POV,C,VAR
event=SendNMEAPort2 PFV,REM,C
#event=StatusMessage Vario Mode

mode=default
type=key
data=S
#event=SendNMEAPort1 POV,C,STF
event=SendNMEAPort2 PFV,REM,S
#event=StatusMessage Speed to Fly Mode

mode=default
type=key
data=A
event=SendNMEAPort2 PFV,REM,A
event=StatusMessage Automatic Mode
```

4. If you use a variant of the stick remote control with a toggle switch, you can skip the remaining menu items. If you have a version with a button, you have to update your stick remote control.

5. How it works, Stefan Langer has published here in a video: https://youtu.be/vwJqPAwFBlU

6. To do this, install the Arduino IDE on your PC (https://www.arduino.cc/en/software)

7. Then download the file bounce.zip in our download area, unzip it and save the unzipped folder in the library directory of the Arduino IDE. Where you can find the library directory depends on the operating system on your computer. It's best to google it.

8. Then download the file Stick_Remote_Control_FV.zip from us and unzip it

9. Opens the Stick_Remote_Control_FV.ino file

10. Click on Tools in the Arduino IDE and select the Arduino Leonardo at Board

11. As the port you choose the one that came last when you connected the stick remote control to the computer via the USB cable

12. Click on “Upload” and that's it

XII. Settings for the stick remote control without FreeVario

If you import a backup of your OpenVario settings, the openvario.xci file will automatically be overwritten with your old version and should be set correctly again. If you use the original openvario.xci file from the image instead, you have to make a few adjustments. To make changes in the openvario.xci file, you must use a text editor that does not do text formatting. I use VSCode for this. The following changes must be made:

```
mode=default
type=key
data=V
#event=SendNMEAPort1 POV,C,VAR
event=SendNMEAPort2 PFV,REM,C
#event=StatusMessage Vario Mode

mode=default
type=key
data=S
#event=SendNMEAPort1 POV,C,STF
event=SendNMEAPort2 PFV,REM,S
#event=StatusMessage Speed to Fly Mode
```

Change to:

```
mode=default
type=key
data=V
event=SendNMEAPort1 POV,C,VAR
#event=SendNMEAPort2 PFV,REM,C
event=StatusMessage Vario Mode

mode=default
type=key
data=S
event=SendNMEAPort1 POV,C,STF
#event=SendNMEAPort2 PFV,REM,S
event=StatusMessage Speed to Fly Mode
```

XIII. Installation of the new OpenVario menu from Kedder

1. The OpenVario must be connected to the Internet and a keyboard must be connected.

2. Start OpenVario and go to the OV menu. Select "Exit to the shell" and confirm with Yes.

3. Enter the following command there:
   - wget https://raw.githubusercontent.com/kedder/openvario-shell/master/scripts/openvario-shell-install.sh -O - | sh

4. Use ESC to return to the main menu and select "Restart" and confirm with Yes. The new menu is now used.

5. If the menu is to be uninstalled again, use the command:
   - wget https://raw.githubusercontent.com/kedder/openvario-shell/master/scripts/openvario-shell-uninstall.sh -O - | sh

6. The menu can be updated via the menu itself or using the shell. The following commands are required in the shell:
   - opkg update
   - opkg upgrade

XIV. Update maps, airspaces etc. with the new menu from Kedder

1. Create a USB stick with your backup as described in Chapter IV, Variant 2

2. Put the stick in your PC and go to the directory /openvario/backup/home/root/.xcsoar

3. Put the files you want to update here

4. Insert the stick into the OpenVario

5. Start the OpenVario and select “Applications” -> “Backup” -> “Restore” from the menu

6. All files that are newer than those on the OpenVario are updated

XV. Using an LTE stick

1. If a Freevario image larger than 22163 is used on the OpenVario, it can go online with an LTE stick (Huawei E3372h-320).

2. The PIN query must be switched off for the SIM card

3. Simply plug the stick into a free USB port and when it is connected to the mobile network, you can go online via LTE and download weather information from PCMet, for example

XVI. Installation



1. USB-C port to update the two ESP's
2. 3.5mm jack socket for external speaker
3. Plug for speed-to-fly switch/flap switch and PTT switch
   a. PTT
   b. STF switch
   c. STF switch
4. Switch which ESP should be updated
5. IGC plug for OpenVario

1. Slide the two hollow screws over the axis of the encoder or potentiometer and screw them to the FreeVario

2. It is essential to shorten (3mm + instrument panel thickness) the two M4 screws before assembly and check for correct length, otherwise the display will be destroyed!

3. Fasten the FreeVario with the M4 screws

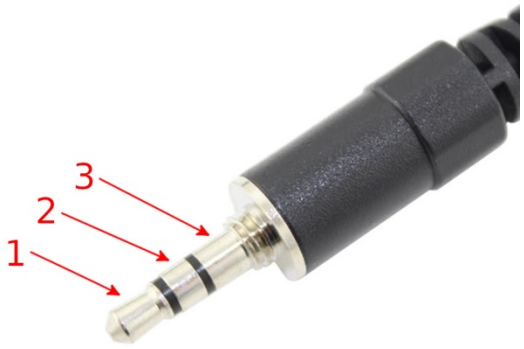
4. Tighten the two rotary knobs. Make sure that the encoder button has enough space to the rear for the push button to work

5. Attach the cover caps to the rotary knobs

6. Connect FreeVario to OpenVario with an IGC-compliant cable. If the FreeVario is connected to OpenVario, a 1:1 cable is used. If it is connected to Larus, a crossover cable must be used.

7. If necessary, connect the flap switch (closed, if the vario is to be set to STF) and PTT button to the three-pin plug

8. To avoid reception problems, the antennas of other devices should be installed as far away as possible from the FreeVario



Bis Hardwareversion V2.0
1. Audio +
2. NC
3. Audio -

Ab Hardwareversion V2.1
1. Audio +
2. Audio -
3. GND

XVII. Disclaimer

The vario gauge is not EASA or FAA certified. Note that this is exactly the same for all the fancy commercial electronic varios and flight computers you are probably used to.

If would you like to know, if you are allowed to use it legally at your glider, ask this question to your airworthiness inspector. Can you legally install any other big brand (as well, non certified) vario gauges? If yes, you should also be able to legally install this vario gauge. If you are under EASA rules see CS-STAN, standard change CS-SC402a.

This instrument is a craft project and not an aviation-approved instrument. I explicitly point out that the use is made under exclusion of any warranty at your own risk!